

“AI-Powered Personalized Diet Consultant”



A

**PBL Report
Submitted to the
S.A.G.E. University, Bhopal, M.P.
in partial fulfillment of the requirements for the award of the
Degree of
BTech (Hons) CSE Specialization in
DA
II Semester
By
Adarsh Kumar Thakur
BTE25DTA100002**

**Under the Guidance of
Prince Narula
Assistant Professor**

**SCHOOL OF COMPUTER TECHNOLOGY
SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY,
BHOPAL**

AUTUMN 2025-26



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CERTIFICATE

This is to certify that the work embodies in this project entitled “**AI-Powered Personalized Diet Consultant**” being submitted by **Adarsh Kumar Thakur [BTE25DTA100002]** in partial fulfillment of the requirement for the award of the degree of **BTech (Hons) CSE – DA** to the School of Computer Technology, Sanjeev Agrawal Global Educational University, Bhopal (M.P) during the academic year **2025-26** is a record of bonafide piece of work, undertaken by him/her under the supervision of the undersigned.

Prince Narula
Assistant Professor
(Project Guide)

Dr. Gourav Shrivastava
I/C HOS



SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL

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CERTIFICATE OF APPROVAL

The Project entitled “**AI-Powered Personalized Diet Consultant**” being submitted by **Adarsh Kumar Thakur [BTE25DTA100002]** has been examined by us and is hereby approved for the award of the degree of **BTech (Hons) CSE - DA**, for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn there in, but approve the project only for the purpose for which it has been submitted.

(Internal Examiner)

(External Examiner)



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DECLARATION

I hereby declare that the work, which is being presented in this project entitled “**AI-Powered Personalized Diet Consultant**” for fulfillment of the requirements for the award of the degree of **BTech (Hons) CSE – AI/CSF/FSD/DA** submitted in the School of Computer Technology, Sanjeev Agrawal Global Educational University, Bhopal, M.P. is an authentic record of my own work carried under the guidance of “**Prince Narula**”. I have not submitted the matter embodied in this report for the award of any other degree.

Adarsh kumar Thakur

BTE25DTA100002



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Adarsh kumar Thakur
BTE25DTA100002

ABSTRACT

The **AI-Powered Personalized Diet Consultant** is an intelligent web application designed to provide personalized nutritional guidance and diet recommendations using Artificial Intelligence. The primary objective of this project is to assist users in maintaining a healthy lifestyle by generating customized meal plans based on their individual health profiles, dietary preferences, and fitness goals.

The system collects essential user information such as age, gender, height, weight, activity level, food preferences, allergies, and health objectives. Using this data, the application calculates Body Mass Index (BMI) and analyzes nutritional requirements. An AI-powered recommendation engine then generates personalized diet plans with detailed calorie and macronutrient distributions.

The application is developed using modern web technologies including **React.js** and **Tailwind CSS** for the frontend, **Flask** for backend development, and **PostgreSQL** for database management. Integration with advanced Large Language Models (LLMs) such as **OpenAI** enables the system to deliver intelligent and dynamic dietary recommendations.

In addition to diet generation, the platform provides features such as user profile management, progress tracking, health goal monitoring, and secure data storage. The system aims to bridge the gap between professional nutritional advice and everyday users by offering affordable, accessible, and personalized dietary assistance.

The project demonstrates how Artificial Intelligence can be effectively integrated with modern web applications to improve healthcare and nutrition management. It provides a scalable solution that can be further enhanced with wearable device integration, food image recognition, multilingual support, and advanced health analytics in the future.

TABLE OF CONTENTS

Title page		
Certificate		i
Certificate of Approval		ii
Declaration		iii
Acknowledgement		iv
Abstract		v
Table of Content		vi
List of Figures		vii
List of Tables		viii
CHAPTER	NAME OF CONTENT	PAGE No.
CHAPTER-1	INTRODUCTION	01-04
1.1	Introduction	
1.2	Project Overview	
1.3	Technology Stack	
CHAPTER-2	LITERATURE SURVEY	
2.1	Overview	
2.2	Existing Systems	
2.3	Problem Statement	
CHAPTER-3	OBJECTIVE & MOTIVATION	
3.1	Objectives	
3.2	Motivation	
CHAPTER-4	PROBLEM IDENTIFICATION& PROPOSED WORK	
4.1	Problems Identified	

4.2	Proposed Solution	
CHAPTER-5	IMPLEMENTATION & RESULT ANALYSIS	
5.1	System Architecture	
5.2	Frontend Implementation	
5.3	Backend Implementation	
5.4	Database Design	
5.5	AI Integration	
5.6	Results	
CHAPTER-6	CONCLUSION AND FUTURE SCOPE	
6.1	Conclusion	
6.2	Future Scope	
	REFERENCES	

LIST OF FIGURES

S.N O	NAME OF FIGURE	PAGE No.
5.1	Home Page of NutriAI	
5.2	User Registration Page	
5.3	AI Assistant	
5.4	User Dashboard	
5.5	User Profile Page	
5.6	Weekly Meal Plan	
5.7	BMI calculations	

CHAPTER-1: INTRODUCTION

1.1 Introduction:

In today's digital era, maintaining a healthy lifestyle has become increasingly difficult due to busy schedules, unhealthy eating habits, and limited access to professional nutritional guidance. Many individuals struggle to select appropriate diets that match their health conditions, fitness goals, and personal preferences. Generic diet plans often fail to address individual nutritional requirements, leading to ineffective results.

Artificial Intelligence (AI) has transformed various industries by providing intelligent and personalized recommendations. In the healthcare and nutrition sector, AI can analyse user-specific information and generate customized diet plans that suit individual needs.

The AI-Powered Personalized Diet Consultant is a web-based application developed to provide customized meal recommendations and nutritional guidance. The system collects user information such as age, gender, height, weight, activity level, dietary preferences, allergies, and health objectives. Based on this information, the application analyzes nutritional requirements and generates personalized diet plans using Artificial Intelligence.

The platform aims to make professional dietary guidance accessible, affordable, and convenient for everyone. By leveraging modern web technologies and AI models, the system helps users make informed dietary decisions and maintain a healthier lifestyle.

1.2 Project Overview:

The AI-Powered Personalized Diet Consultant is a full-stack web application that combines Artificial Intelligence with modern web development technologies. The platform enables users to create personal health profiles and receive customized diet recommendations based on their unique requirements.

The application offers several important features, including:

- User Registration and Authentication

- **Personalized User Profile Management**
- **BMI Calculation**
- **AI-Generated Diet Recommendations**
- **Daily and Weekly Meal Planning**
- **Nutritional Analysis**
- **Progress Tracking**
- **Secure Data Storage**

The system uses AI models to analyze user data and generate meal plans that align with the user's health objectives. By automating the diet planning process, the application helps users save time while making informed nutritional decisions.

1.3 Technology Stack

The AI-Powered Personalized Diet Consultant is developed using the following technologies:

Frontend Technologies:

- React.js
- Tailwind CSS
- JavaScript
- HTML5
- CSS3

Backend Technologies:

- Python
- Flask Framework

Database:

- PostgreSQL

Artificial Intelligence:

- OpenAI API / Google Gemini API

Deployment Platform:

- Render Cloud Platform

These technologies work together to provide a responsive user interface, secure data management, efficient backend processing, and intelligent diet recommendation capabilities.

CHAPTER-2: LITERATURE SURVEY

2.1 Overview

The increasing awareness of health and fitness has created a growing demand for personalized nutritional guidance. Proper nutrition plays a vital role in maintaining overall health, improving physical performance, and preventing lifestyle-related diseases. However, creating an effective diet plan requires knowledge of nutritional science and individual health requirements.

With advancements in Artificial Intelligence (AI), healthcare and nutrition services are becoming more intelligent and personalized. AI-based systems can analyse user data, identify dietary needs, and generate customized recommendations. These technologies help users make informed decisions about their food intake and maintain healthier lifestyles.

The AI-Powered Personalized Diet Consultant utilizes Artificial Intelligence and modern web technologies to provide personalized diet recommendations based on user-specific information such as age, weight, height, activity level, dietary preferences, and fitness goals.

2.2 Existing Systems

Several diet and nutrition applications are currently available in the market. Popular platforms such as MyFitnessPal, Healthily, Fitbit, and Nutrify Me provide features including calorie tracking, meal logging, and fitness monitoring.

Common features available in existing systems include:

- Calorie counting and food tracking.
- Exercise and activity monitoring.
- Weight management support.
- Nutritional information for food items.
- Health and fitness goal tracking.

Although these applications are useful, many of them depend on predefined meal plans and limited personalization. Users often receive generic recommendations that may not fully satisfy their individual nutritional requirements.

Additionally, some platforms require paid subscriptions to access advanced features, making professional dietary guidance less accessible for many users.

2.3 Problem Statement

Traditional diet planning methods often fail to consider the unique requirements of individual users. Factors such as age, gender, body weight, activity level, allergies, medical conditions, and food preferences significantly influence nutritional needs.

The major problems identified are:

- Lack of personalized diet recommendations.
- Limited access to professional nutrition experts.
- Generic diet plans that may not suit every individual.
- Difficulty in calculating nutritional requirements.
- Lack of continuous monitoring and progress tracking.
- High consultation costs associated with professional dieticians.

To overcome these challenges, there is a need for an intelligent system capable of generating personalized diet plans automatically. The AI-Powered Personalized Diet Consultant addresses these issues by using Artificial Intelligence to analyse user data and generate customized meal recommendations that support healthier lifestyle choices.

CHAPTER-3: OBJECTIVES & MOTIVATION

3.1 Objectives

The primary objective of the AI-Powered Personalized Diet Consultant is to provide intelligent and personalized nutritional guidance using Artificial Intelligence. The system is designed to help users maintain healthy eating habits and achieve their fitness goals through customized diet recommendations.

The main objectives of the project are:

- To develop a user-friendly web application for personalized diet consultation.
- To collect and manage user health information such as age, weight, height, gender, and activity level.
- To calculate Body Mass Index (BMI) and assess the user's health condition.
- To generate personalized diet plans using Artificial Intelligence.
- To provide calorie and nutritional information for recommended meals.
- To track user progress and support health goal monitoring.
- To store user information securely using PostgreSQL database.
- To improve accessibility to dietary guidance through a digital platform.
- To integrate modern web technologies and AI for efficient decision-making.
- To encourage healthier lifestyles through personalized nutrition recommendations.

3.2 Motivation

The motivation behind developing the AI-Powered Personalized Diet Consultant comes from the increasing number of health problems caused by unhealthy eating habits and poor nutritional awareness. Many individuals face difficulties in

selecting appropriate diets due to lack of knowledge, busy schedules, and limited access to professional dieticians.

Traditional diet plans are often generic and may not suit the unique requirements of every individual. Factors such as age, weight, fitness goals, allergies, and dietary preferences vary from person to person, making personalized nutrition essential for achieving effective results.

Artificial Intelligence provides an opportunity to automate and improve the diet planning process by analysing user-specific data and generating customized recommendations. This project aims to make professional dietary guidance more accessible, affordable, and convenient for everyone.

The project also serves as an opportunity to apply knowledge of web development, database management, and Artificial Intelligence in solving a real-world healthcare problem. By combining technology with nutrition science, the system contributes toward improving health awareness and promoting better lifestyle choices among users.

CHAPTER-4: PROBLEM IDENTIFICATION & SOLUTION

4.1 Problems Identified

During the study of existing diet and nutrition platforms, several challenges were identified that affect users seeking proper dietary guidance and healthy lifestyle management.

Problem 1: Lack of Personalized Diet Plans

Most traditional diet plans are generic and designed for a broad audience. They do not consider individual factors such as age, gender, body weight, activity level, allergies, and health goals. As a result, users often fail to achieve desired outcomes.

Problem 2: Limited Access to Nutrition Experts

Professional dietitians and nutrition consultants may not be easily accessible to everyone. Many individuals cannot afford regular consultations due to high costs or geographical limitations.

Problem 3: Difficulty in Nutritional Analysis

Calculating daily calorie requirements, protein intake, carbohydrates, fats, and other nutritional values can be complex for ordinary users. This often leads to unhealthy food choices.

Problem 4: Lack of Progress Monitoring

Many people start a diet plan but fail to track their progress effectively. Without proper monitoring, users may lose motivation and discontinue their health journey.

Problem 5: Time-Consuming Diet Planning

Preparing customized meal plans manually requires significant time and nutritional knowledge. Busy individuals often struggle to create balanced diets that meet their health requirements.

Problem 6: Increasing Lifestyle-Related Diseases

Poor eating habits and unhealthy lifestyles contribute to obesity, diabetes, hypertension, and other chronic diseases. There is a growing need for accessible tools that promote healthier eating habits.

4.2 Proposed Solution

To overcome the identified challenges, the AI-Powered Personalized Diet Consultant provides an intelligent and automated solution using Artificial Intelligence and modern web technologies.

Solution 1: Personalized User Profiling

The system collects user information including age, gender, height, weight, activity level, dietary preferences, allergies, and fitness goals. This information is used to create a personalized health profile.

Solution 2: AI-Based Diet Recommendation

The application integrates OpenAI/Gemini APIs to analyse user data and generate customized meal plans. The recommendations are tailored to individual nutritional requirements and health objectives.

Solution 3: BMI and Nutritional Analysis

The system automatically calculates Body Mass Index (BMI) and estimates calorie requirements. It also provides nutritional breakdowns to help users understand their dietary intake.

Solution 4: Progress Tracking

Users can monitor their health progress over time by tracking weight changes, dietary adherence, and fitness goals. This helps maintain motivation and encourages consistency.

Solution 5: Secure Data Management

All user information is securely stored in a PostgreSQL database. The system ensures reliable data management and easy access to historical records.

Solution 6: User-Friendly Interface

The application is developed using React.js and Tailwind CSS, providing a responsive and intuitive user experience across desktops, tablets, and mobile devices.

By combining Artificial Intelligence with modern web technologies, the AI-Powered Personalized Diet Consultant offers an efficient, accessible, and scalable solution for personalized nutrition and health management.

CHAPTER-5: IMPLEMENTATION

5.1 System Architecture

The AI-Powered Personalized Diet Consultant follows a modern three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

The user interacts with the React.js frontend through a web browser. The frontend communicates with Flask REST APIs, which process requests, perform authentication, retrieve user data, and generate personalized recommendations using Artificial Intelligence. PostgreSQL is used to store user profiles, meal plans, BMI records, and progress data.

System Flow:

User → React Frontend → Flask Backend → PostgreSQL Database → AI Engine (Gemini/OpenAI) → Personalized Response

This architecture ensures scalability, security, maintainability, and efficient data management.

5.2 Frontend Implementation

The frontend is developed using React.js and Tailwind CSS to provide a responsive and interactive user experience.

Landing Page

The landing page introduces the NutriAI platform and highlights its major features such as AI-powered nutrition recommendations, calorie tracking, and personalized meal planning.

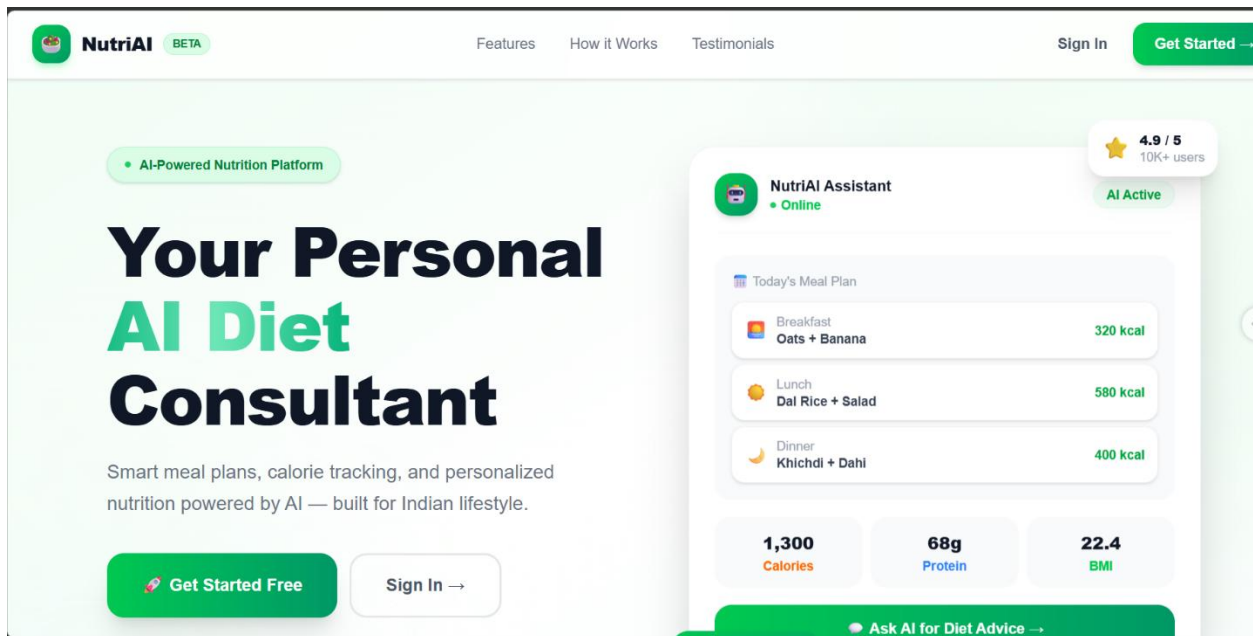


Figure 5.1: Home Page of NutriAI

Multi-Step Registration System

A guided registration process is implemented to collect detailed user information in multiple stages.

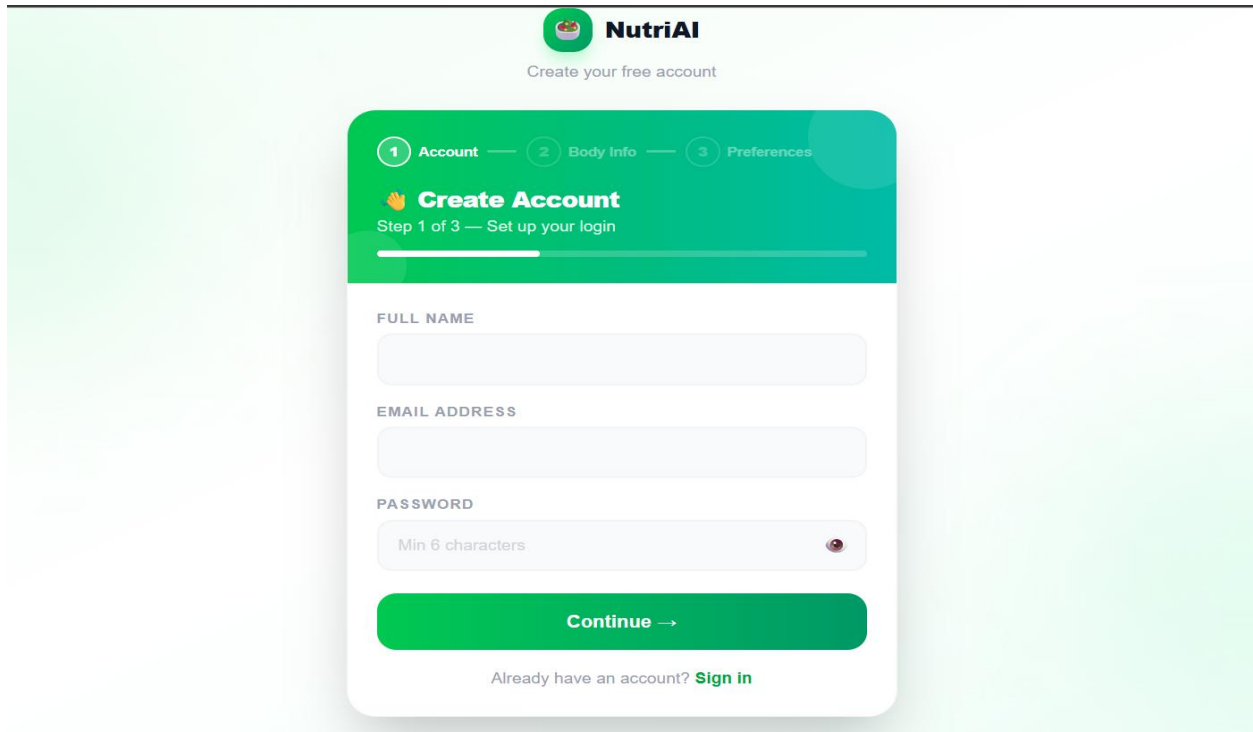


Figure 5.2: User Registration Page

5.3 Backend Implementation

The backend is developed using Python Flask.

Major backend functionalities include:

- User Authentication
- Session Management
- API Development
- Profile Management
- BMI Calculation
- Meal Plan Generation
- AI Recommendation Processing
- Database Operations

The backend acts as a communication bridge between the frontend, PostgreSQL database, and AI engine.

5.4 Database Design

PostgreSQL is used for secure storage and management of application data.

The database contains multiple tables including:

and future scalability.

5.5 AI Integration

Artificial Intelligence is the core component of the system.

The application integrates Gemini/OpenAI APIs to generate personalized nutritional guidance.

The AI model analyzes:

- Age
- Weight
- Height
- BMI
- Activity Level
- Dietary Preferences
- Health Goals

Based on these inputs, the AI generates personalized recommendations and meal plans.

AI Assistant Module

A dedicated AI Chat Assistant allows users to ask nutrition-related questions and receive intelligent responses.

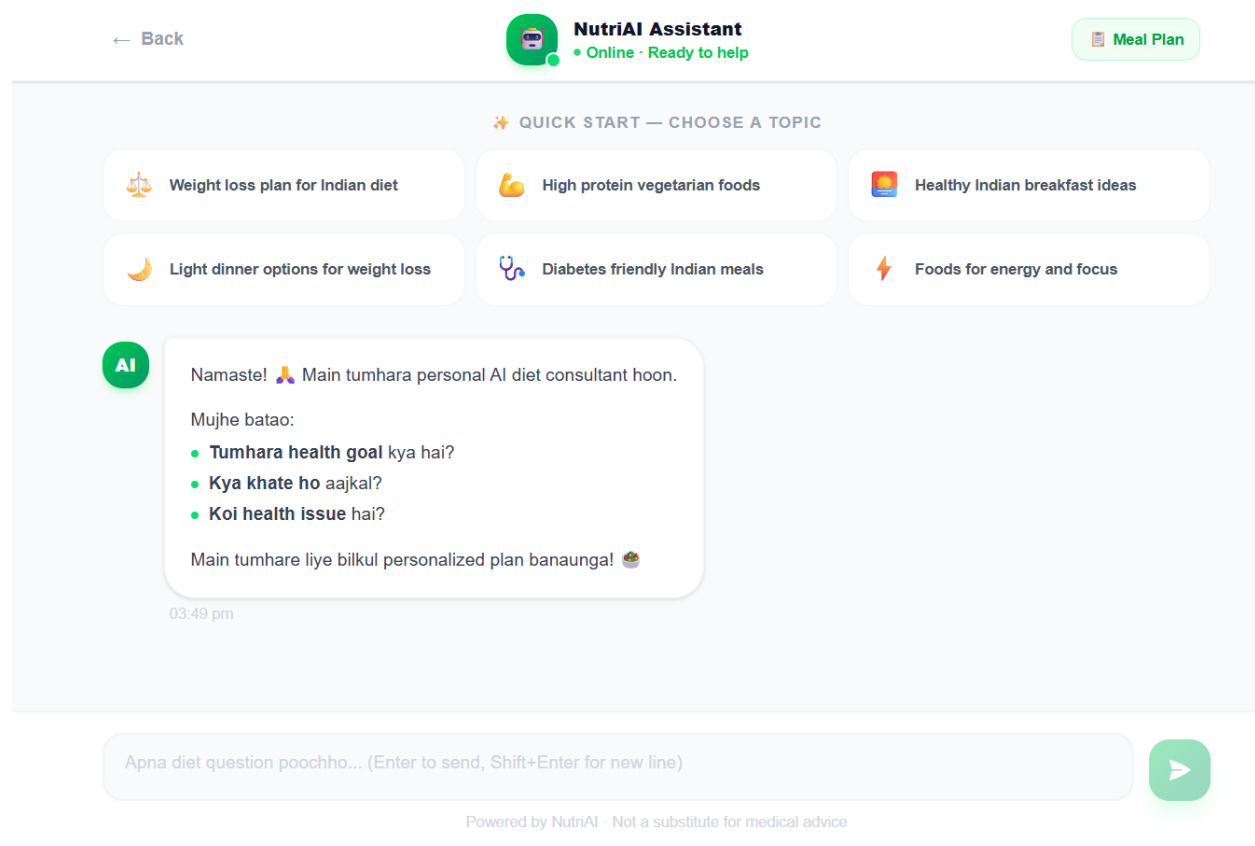


Figure 5.3: AI Assistant

5.6 Results

The developed application successfully implemented all major project objectives.

Dashboard Module

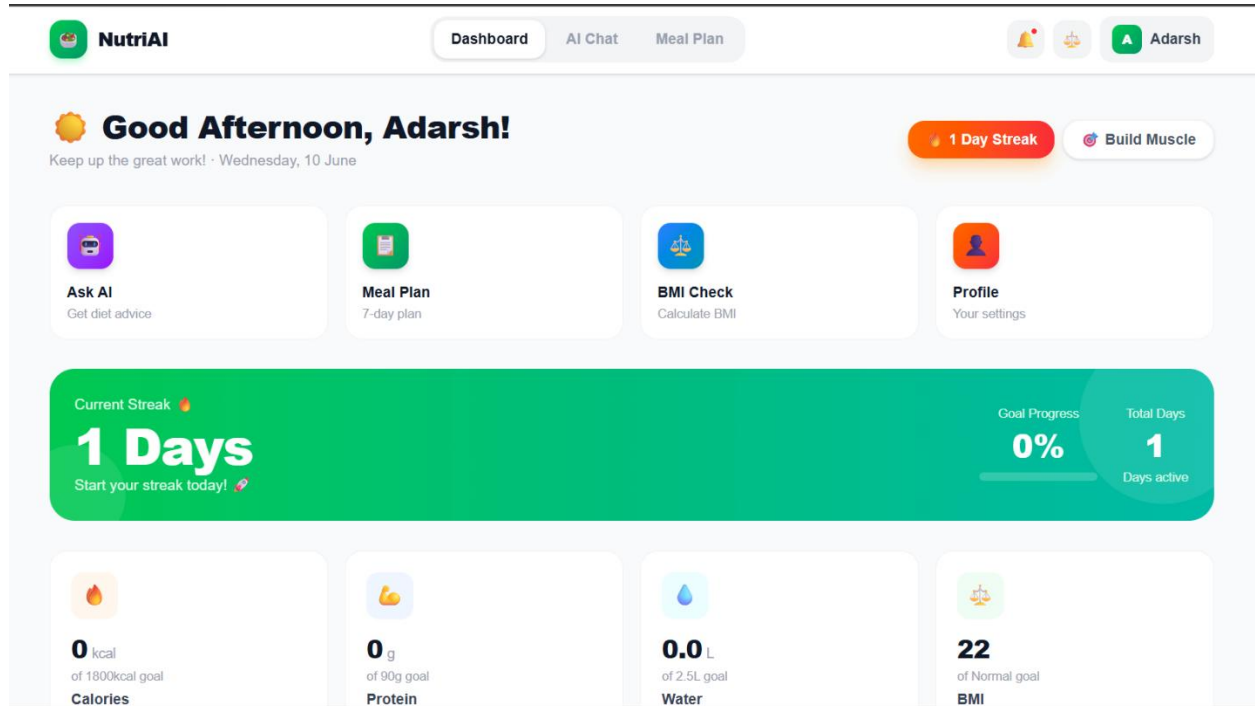


Figure 5.4: User Dashboard

Profile Management Module

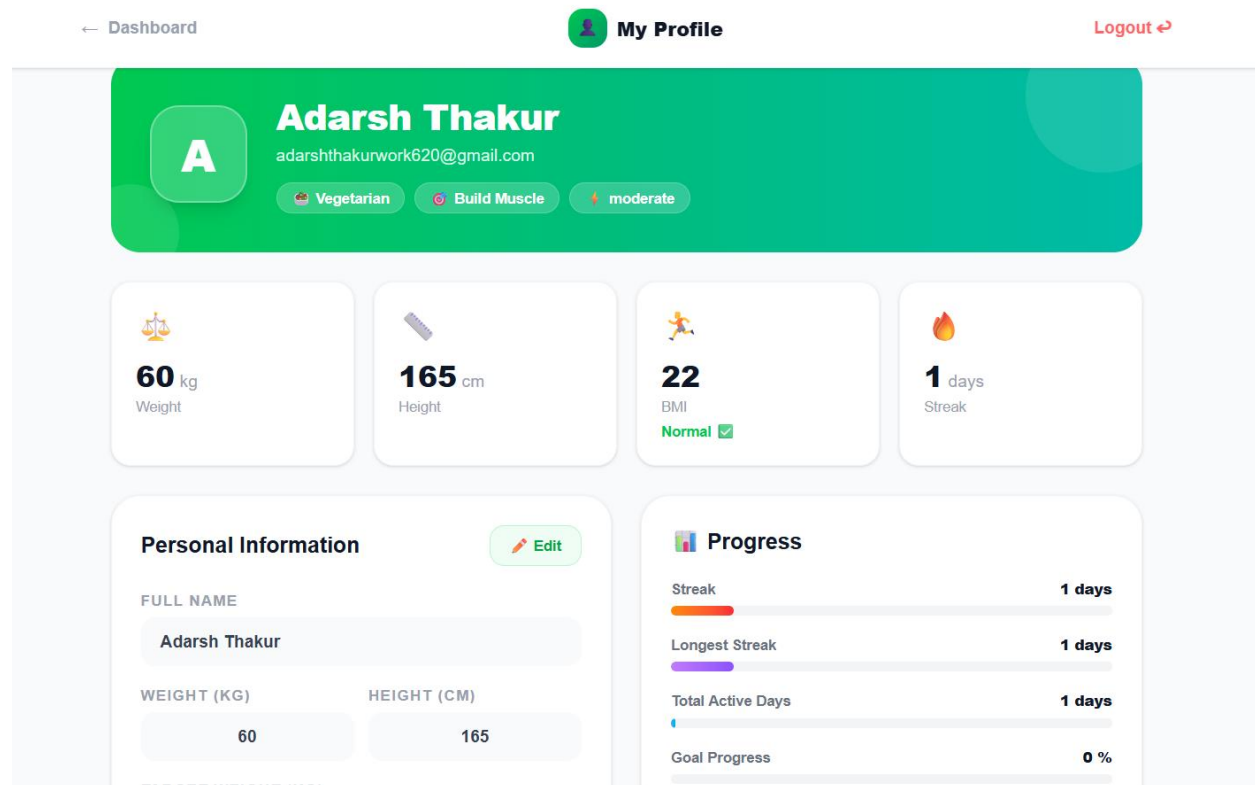


Figure 5.5: User Profile Page

Weekly Meal Plan Module

The system generates a complete weekly meal schedule based on the user's health profile and fitness goal.

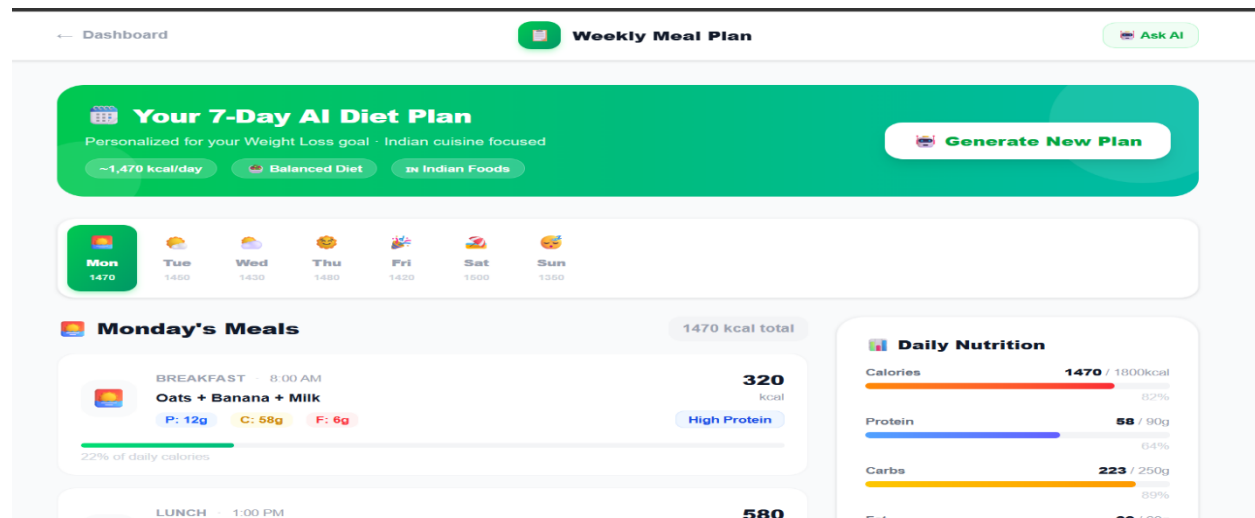


Figure 5.6: Weekly Meal Plan

AI Assistant Module

The AI Assistant successfully generates personalized nutritional guidance within seconds and provides contextual responses according to user requirements.

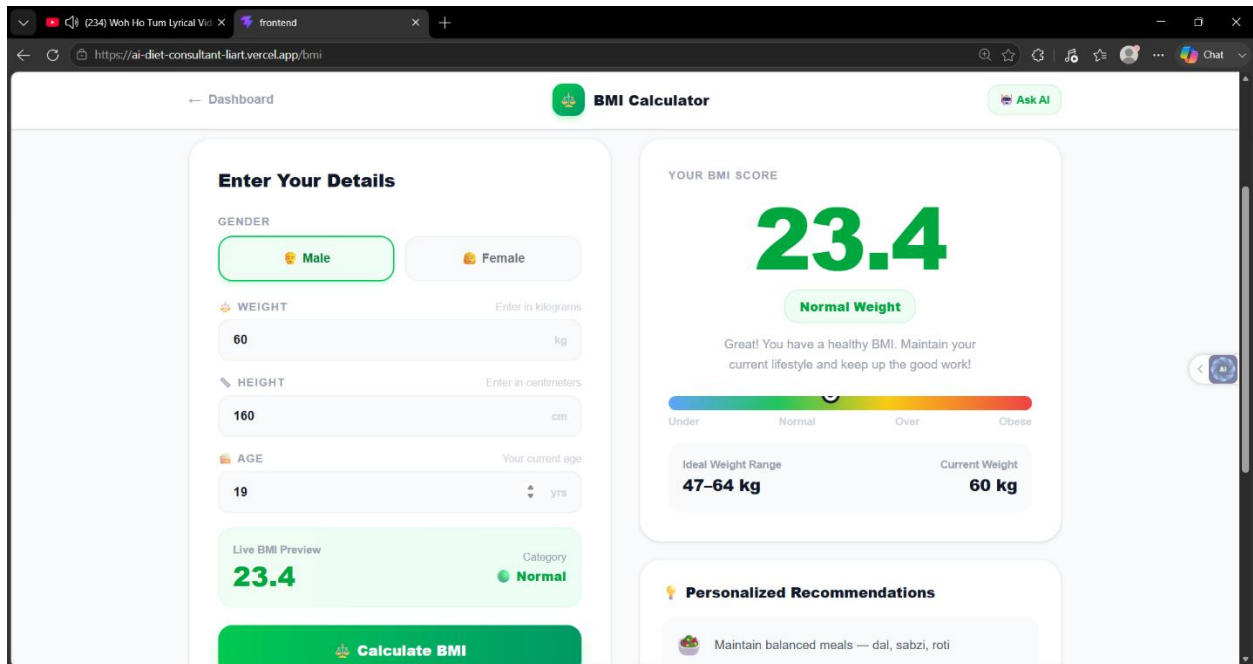


Figure 5.7: BMI calculations

CHAPTER-6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The **AI-Powered Personalized Diet Consultant (NutriAI)** was developed to provide intelligent and personalized nutritional guidance using Artificial Intelligence and modern web technologies. The system successfully combines user profiling, BMI analysis, AI-powered diet recommendations, weekly meal planning, and health progress tracking within a single platform.

The application enables users to create personalized health profiles and receive customized meal plans based on their dietary preferences, activity levels, and fitness goals. Features such as the AI Assistant, Weekly Meal Planner, BMI Calculator, Dashboard Analytics, and Progress Tracking enhance the overall user experience and make nutrition management more accessible and effective.

The project was implemented using React.js for the frontend, Flask for backend development, PostgreSQL for database management, and Gemini/OpenAI APIs for intelligent recommendation generation. The integration of these technologies resulted in a responsive, scalable, and user-friendly application.

The developed system successfully achieved the objectives of providing personalized dietary guidance, improving nutritional awareness, and helping users make informed health decisions. The project demonstrates how Artificial Intelligence can be effectively applied in the healthcare and nutrition domain to solve real-world problems.

6.2 Future Scope

Although the current system provides comprehensive diet consultation features, several enhancements can be implemented in future versions to improve functionality and user experience.

Future improvements may include:

- Mobile application development for Android and iOS platforms.
- Integration with wearable fitness devices such as smartwatches and fitness bands.
- Food image recognition for automatic calorie estimation using Computer Vision.
- Voice-enabled AI Nutrition Assistant for hands-free interaction.
- Real-time health monitoring and analytics dashboard.
- Personalized workout recommendations along with diet plans.
- Integration with healthcare professionals and certified dietitians.
- Multilingual support for regional and international users.
- Advanced nutrient deficiency detection and health risk prediction.
- Cloud-based data synchronization and backup.
- AI-powered grocery list generation based on weekly meal plans.
- Personalized recipe recommendations according to user preferences.

By incorporating these enhancements, the AI-Powered Personalized Diet Consultant can evolve into a complete digital health and nutrition ecosystem, offering smarter, more accurate, and highly personalized wellness solutions for users worldwide.

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